

Retrieval-Augmented Generation (RAG)

What is RAG?

Retrieval-Augmented Generation (RAG) is an AI architecture that combines:

1. **Retrieval** – Fetching relevant information from external knowledge sources.
2. **Augmentation** – Adding the retrieved information to the user's prompt as additional context.
3. **Generation** – Using an LLM to generate an accurate answer based on both the user's question and the retrieved context.

Instead of relying only on what the LLM learned during training, RAG allows it to use **up-to-date, private, and domain-specific knowledge**.

Why do we need RAG?

Large Language Models (LLMs) like GPT, Claude, Gemini, or Llama are trained on massive datasets.

However, they have several limitations:

- They cannot access your company's private documents.
- They don't automatically know newly created information.
- They may generate incorrect answers (hallucinations).
- Retraining or fine-tuning for every new document is expensive and time-consuming.

RAG solves these problems by retrieving relevant information at inference time.

Understanding the Three Components

1. Retrieval

Retrieve the most relevant information from external sources.

These sources may include:

- PDFs
- Word documents
- Excel files
- Databases

- SharePoint
- Confluence
- Websites
- APIs
- Emails
- Knowledge bases

Example:

User asks:

Why is my Vodafone bill £2,500 this month?

Instead of guessing, the system retrieves:

- Customer billing history
- Current month's bill
- International call records
- Roaming usage
- Customer account details

2. Augmentation

The retrieved information is added to the prompt.

Example prompt sent to the LLM:

User Question:

Why is my bill £2,500?

Retrieved Context:

Customer Name: John

Previous Bills:

£120

£140

£135

Current Usage:

27 International Calls

350 International SMS

Roaming Charges £1,950



Answer the question using only this information.

Now the LLM has factual evidence.

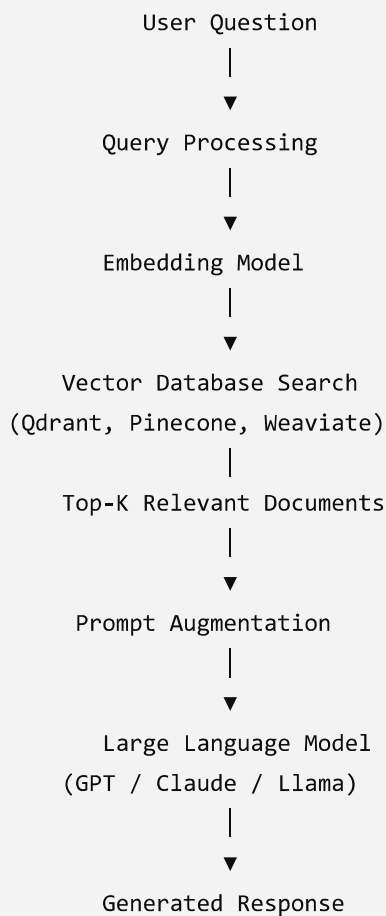
3. Generation

Finally, the LLM generates an answer using the retrieved context.

Example:

Your bill increased because you made 27 international calls while roaming abroad, resulting in £1,950 in roaming charges. Your previous bills were around £120–£140, making this month's bill significantly higher.

Complete RAG Architecture



Real-Life Vodafone Example

Without RAG

User:

Why is my Vodafone bill £2,500?

ChatGPT:

 **ChatGPT** It might be because of roaming, taxes, late payment or international calls.  

This is only a generic guess.

With RAG

The system retrieves:

Customer Database

Current Bill

Call Logs

Usage Details

International Charges

Roaming Charges

LLM Response:

Your bill is higher because you travelled to Germany from 5th–12th July and incurred £1,950 in roaming charges along with 27 international calls and 350 SMS.

This answer is accurate because it uses actual customer data.

Problems Solved by RAG

1. Hallucination Reduction

Without RAG

LLM guesses.

With RAG

LLM answers using retrieved evidence.

2. Access to Private Data

Examples:

- Company policies
- HR documents
- Banking records
- Medical reports
- Legal contracts
- Customer databases

The LLM never had these during training.

RAG provides access securely.

3. Up-to-Date Knowledge

Instead of retraining the model every week:

- Add new PDFs
- Update Confluence
- Insert database records

RAG retrieves the latest information immediately.

4. Lower Cost

Fine-tuning costs time and money.

RAG simply indexes new documents.

5. Explainability

Since the answer comes from retrieved documents, the system can cite its sources.

Example:

Source:

Billing_Report_July.pdf

Customer_DB_Record

Common RAG Use Cases

Customer Support Chatbots

- Telecom
- Banking
- Insurance
- Airlines

Enterprise Knowledge Assistant

Search:

- HR policies
- Leave policy
- Company handbook
- Internal documentation

AI Resume Coach

Upload a resume.

Ask:

- What are my strengths?
 - Which skills are missing?
 - Suggest interview questions.
-

Medical Assistant

Retrieve:

- Patient history
- Medical reports
- Clinical guidelines

Generate evidence-based recommendations.

Legal Assistant

Search through:

- Contracts
- Agreements
- Regulations

Answer legal queries based on the retrieved clauses.

Financial Assistant

Retrieve:

- Transactions
- Bank statements
- Loan records

Provide personalised financial insights.

Learning Assistant

Upload:

- Lecture notes
- PDFs
- Course materials
- Video transcripts

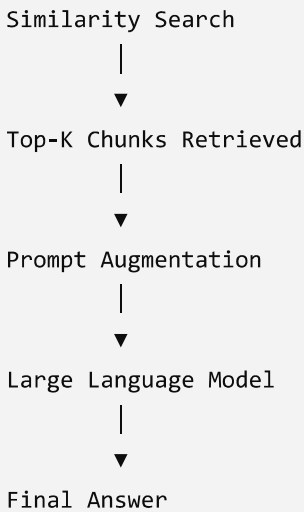
Students can ask questions and receive answers grounded in those materials.

RAG vs Fine-Tuning

Feature	RAG	Fine-Tuning
Uses private documents	✓	✓
Requires model retraining	✗	✓
Easy to update	✓	✗
Lower cost	✓	✗
Reduces hallucinations	✓	Partial
Keeps knowledge current	✓	✗
Works with databases	✓	✗

Typical RAG Pipeline





Interview Answer (2-Minute Version)

Retrieval-Augmented Generation (RAG) is an AI architecture that enhances Large Language Models by combining information retrieval with text generation. Instead of relying only on the model's training data, RAG retrieves relevant information from external sources such as PDFs, databases, Confluence, SharePoint, APIs, or knowledge bases. The retrieved content is added to the user's prompt, and the LLM generates an answer grounded in that context.

RAG addresses several key challenges of LLMs: it reduces hallucinations, enables access to private and organisation-specific data, keeps responses current without retraining the model, and is significantly more cost-effective than frequent fine-tuning. A typical RAG pipeline involves document ingestion, chunking, embedding generation, storage in a vector database, similarity search, prompt augmentation, and response generation using an LLM.

Common use cases include enterprise knowledge assistants, customer support chatbots, banking and healthcare assistants, legal document search, AI resume coaches, and educational question-answering systems.

